

Micro Risks and (Robust) Pareto Improving Policies

Mark Aguiar, Manuel Amador, Cristina Arellano

Presenter: Shi Hu
BGGMP Reading Group, Fall 2022

Sep 21, 2022

Introduction

- many (advanced) economies often experience low interest rates $r < g$ (Mehrotra-Sergeyev 2020)
- what is the role of government debt in such environment?
- what could be achieved through fiscal policy in such environment?

- fiscal policy in Bewley-Huggett-Aiyagari economy
 - incompleteness gives incentives to save $\rightarrow$ low rates
 - allow very general heterogeneity
 - allow wedge between F_K and $r + \delta$
- Robust Pareto Improvements (RPI)
 - very limited knowledge of individual preferences required (robust!)
 - make every agent better off in every state and every date
- feasibility with limited fiscal instruments
 - simply policies: lump-sum transfers, linear taxes, government debt
 - key: elasticity of aggregate savings

Households

- no population and technology growth (easy to add back)
 $r < g \Leftrightarrow r < 0$
- allow **very general** preference heterogeneity
 - no wealth effect¹ (for example, GHH)
 - (individual and aggregate) labor supply only depends on wage w_t
- HHs $i \in [0, 1]$ with idiosyncratic labor productivity z_t^i and managerial/entrepreneurial talent θ_t^i
 - $\int \theta_t^i di = 1$ for any t
 - BC: $c_t^i + a_{t+1}^i \leq w_t z_t^i n_t^i + \theta_t^i \Pi_t + (1 + r_t) a_t^i + T_t$
 - save in risk-free asset, with borrowing constraint $a_{t+1}^i \geq \underline{a}$

¹for simplicity purpose. Analysis could be extended to preferences with wealth effects, in which the elasticity of aggregate labor supply will also matter.

- representative firm, CRS technology $F(K, L)$
 - constant markup $\mu \geq 1$
 - purchases factors competitively: $(1 + \tau^k)(r + \delta)$ and $(1 + \tau^n)w$
 - FOC: $F_K(K, L) = \mu(1 + \tau^k)(r + \delta)$ and $F_L(K, L) = \mu(1 + \tau^n)w$
 - pre-tax profit $\hat{\Pi} = \frac{\mu-1}{\mu} F(K, L)$
 - post-tax profit $\Pi = (1 - \tau^\pi) \hat{\Pi}$
- Golden Rule capital level: K^* with $F_K(K, L) = \delta$, for a given L
 - at K^* , aggregate consumption is maximized

- issue bonds $\{B_t\}$
- linear taxes on firms $\{\tau_t^n, \tau_t^k, \tau_t^\pi\}$
- lump-sum transfers $\{T_t\}$
- BC:
$$T_t \leq \underbrace{\tau_t^n w_t N_t + \tau_t^k (r_t + \delta) K_t + \tau_t^\pi \hat{\Pi}_t}_{\text{tax revenue}} + \underbrace{B_{t+1} - (1 + r_t) B_t}_{\text{seigniorage revenue}}$$

Equilibrium

- Given a fiscal sequence $\{B_t, T_t, \tau_t^n, \tau_t^k, \tau_t^\pi\}$, and an initial distribution of HHs $\{a_0^i, z_0^i, \theta_0^i\}$, an **equilibrium** is $\{A_t, K_t, N_t, \Pi_t\}$ and $\{r_t, w_t\}$ such that
 - HHs and firms optimize
 - transfers satisfy the sequential government BC
 - markets clear

$$\text{Asset Market: } A_t = K_t + B_t$$

$$\text{Labor Market: } N_t = L_t$$

$$\text{Goods Market: } C_t = F(K_t, N_t) - K_{t+1} + (1 - \delta)K_t$$

Robust Pareto Improvements (RPI)

- HHs BC: $c_t^i + a_{t+1}^i \leq w_t z_t^i n_t^i + \theta_t^i \Pi_t + (1 + r_t) a_t^i + T_t$
- let $\{w_t^o, r_t^o, \Pi_t^o, T_t^o\}$ be the reference equilibrium
- $\{w_t, r_t, \Pi_t, T_t\}$ RPI if it expands budget set for **every agent** at **every state** and **every date**²
 - $w_t \geq w_t^o$
 - $r_t \geq r_t^o$
 - $\Pi_t \geq \Pi_t^o$
 - $T_t \geq T_t^o - (r_t - r_t^o) \underline{a}$, where $\underline{a} \equiv \inf \underline{a}^i$
 - at least one strict inequality

²in particular, not from a “time-0” or utilitarian perspective

- robust to
 - general preference: could vary across HHs, allow recursive utility
 - intertemporal/inter-state MRS
 - idiosyncratic risks structure
- pro: very limited information needed at micro level
- (potential) con: sounds too good to be true. really achievable?

- robust to
 - general preference: could vary across HHs, allow recursive utility
 - intertemporal/inter-state MRS
 - idiosyncratic risks structure
- pro: very limited information needed at micro level
- (potential) con: sounds too good to be true. really achievable?
 - feasibility of RPI policy

- start from some stationary equilibrium (could relax)
 - $(B^o, \tau^{no}, \tau^{ko}, \tau^{\pi o}, T^o)$ initial fiscal policy
 - $(w^o, r^o, \Pi^o), (N^o, K^o)$ pinned down in the initial equilibrium
 - for simplicity, assume $T^o = 0$
 - so tax revenue equals to $r^o B^o$
- at $t = 0$, announce new policy $\{B_{t+1}, \tau_t^n, \tau_t^k, \tau_t^\pi, T_t\}_{t \geq 0}$
 - after the announcement, perfect foresight
 - HHs and firms re-optimize
 - generate $\{A_t, K_t, N_t, \Pi_t\}, \{r_t, w_t\}$
 - $A_0 = A^o, K_0 = K^o, B_0 = B^o, r_0 = r^o$

- focus on “constant wage and profit policy”: $w_t = w^o, \Pi_t = \Pi^o \forall t$
 - doable via $\{\tau_t^n, \tau_t^\pi\}$
 - recall HHs BC: $c_t^i + a_{t+1}^i \leq w_t z_t^i n_t^i + \theta_t^i \Pi_t + (1 + r_t) a_t^i + T_t$
 - no need to worry about (z_t^i, θ_t^i)
 - HHs affected only through $(\mathbf{r}, \mathbf{T}) \equiv \{r_t, T_t\}_{t \geq 0}$
- $\mathcal{A}_{t+1}(\mathbf{r}, \mathbf{T})$ aggregate HH assets at the end of period t
- aggregate consumption

$$\mathcal{C}_t(\mathbf{r}, \mathbf{T}) = w^o N^o + \Pi^o + (1 + r_t) \mathcal{A}_t(\mathbf{r}, \mathbf{T}) - \mathcal{A}_{t+1}(\mathbf{r}, \mathbf{T}) + T_t$$

Feasibility Condition

- $(\mathbf{r}, \mathbf{T})$ is feasible if $\exists$ fiscal policy $\{B_t, \tau_t^n, \tau_t^k, \tau_t^\pi, T_t\}_{t \geq 0}$ that could implement $\{A_t, K_t, N^o, \Pi^o\}_{t \geq 0}$ and $\{r_t, w^o\}$ in CE with $K_0 = K^o, A_0 = A^o, B_0 = B^o$.

Lemma 1

$(\mathbf{r}, \mathbf{T})$ is feasible *if and only if* $\exists \{K_t\}_{t \geq 0}$ and $\{B_t\}_{t \geq 0}$ with $K_0 = K^o, B_0 = B^o$ such that $\forall t$:

① $\mathcal{A}_{t+1}(\mathbf{r}, \mathbf{T}) = B_{t+1} + K_{t+1}$

② $B_{t+1} - (1 + r_t)B_t - T_t \geq$

$$F(K^o, N^o) - F(K_t, N^o) - (r^o + \delta)K^o + (r_t + \delta)K_t - r^o B^o$$

derive

- **Remark:** feasibility only depends on aggregates

Equivalent Condition

- $C_t = w^o N^o + \Pi^o + (1 + r_t) \mathcal{A}_t + T_t - \mathcal{A}_{t+1}$
- replace government BC with aggregate resource constraint

Corollary 1

$(\mathbf{r}, \mathbf{T})$ is feasible *if and only if* $\exists \{K_t\}$ with $K_0 = K^o$ such that $\forall t$:

$$C_t(\mathbf{r}, \mathbf{T}) \leq F(K_t, N^o) + (1 - \delta)K_t - K_{t+1}$$

derive

Constant- K Policy

- benchmark: starting from laissez-faire and HHs-no-borrowing
stationary equilibrium: no taxes, $T^o = 0$, $B^o = 0$, $\underline{a} = 0$
- consider a new policy through which
 - $r_t > r^o$, $K_t = K^o \forall t \geq 1$
 - $T_t = -T^o - (r_t - r^o)\underline{a} = 0$ (consistent with RPI)
 - feasible $\Leftrightarrow B_{t+1} - (1 + r_t)B_t \geq (r_t - r^o)K^o$
 - $(1 + \tau_t^k)\mu(r_t + \delta) = F_K(K_t, N^o) = F_K(K^o, N^o) = \mu(r^o + \delta) \Rightarrow \tau_t^k = \frac{r^o - r_t}{r_t + \delta} < 0$
- consider the new stationary equilibrium
 - $A' < \infty$, $r' > r^o$, $B' = A' - K^o > 0$
 - feasibility condition: $-r'B' \geq (r' - r^o)K^o$

Constant- K Policy: Steady State

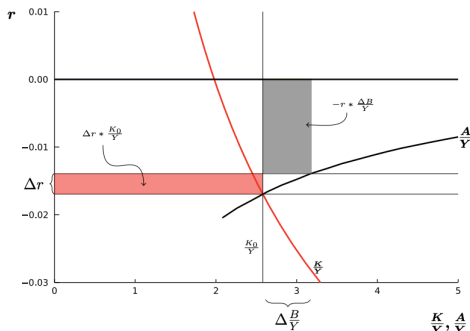


Figure: à la Aiyagari (1994)

- feasible $\Leftrightarrow -r'B' \geq (r' - r^o)K^o \Leftrightarrow \text{gray} > \text{red}$
- the slope of $\frac{A}{Y}$ (elasticity of aggregate saving) matters

Over-accumulated Capital

- capital above Golden Rule: $F_K(K^o, N^o) < \delta$
- **idea:** replace capital with government debt (Diamond 1965)
- consider $(\mathbf{r}, \mathbf{T}) = (r^o, T^o)$, and $\{K_t\}$ with $K_0 = K^o, K_t = K^* \forall t \geq 1$
 - $\mathcal{C}_t(\mathbf{r}, \mathbf{T}) = C^o = F(K^o, N^o) - \delta K^o$
 - $\mathcal{C}_t(\mathbf{r}, \mathbf{T}) < F(K_t, N^o) - K_{t+1} + (1 - \delta)K_t \forall t \Rightarrow$ feasible
 - additional resources everywhere $\Rightarrow$ lump-sum rebate the surplus
 - under some continuity condition on $\mathcal{C}_t$, an implementable RPI exists

detail

Under-accumulated Capital

- starting from $K^o < K^*$ (assuming $\underline{a} = 0$)
- consider $(\mathbf{r}, \mathbf{T}^o)$ with $r_\tau > r^o$ for some $\tau \geq 1$, and $r_t = r^o$ for $t \neq \tau$
- elasticity of aggregate savings at each t : $\xi_{t,\tau} \equiv \frac{\partial \mathcal{A}_t}{\partial r_\tau} \frac{R^o}{\mathcal{A}^o}$
 - $R^o \equiv 1 + r^o$
 - $R_k \equiv 1 + F_K(K^o, N^o) - \delta > 1$
- **sufficient** condition for feasibility:

$$\left(\frac{R_k - R^o}{R^o} \right) \sum_{t=1}^{\infty} R_k^{-(t-\tau)} \xi_{t,\tau} > 1$$

Sufficient Condition

$$\left(\frac{R_k - R^o}{R^o} \right) \sum_{t=1}^{\infty} R_k^{-(t-\tau)} \xi_{t,\tau} > 1$$

- discounted sum of aggregate savings elasticity has to be *large enough*
 - extends the constant- K steady state analysis
- R_k , intertemporal price at which government trades with firms
- R^o , intertemporal price at which government trades with HHs
- $R_k - R^o = (\mu - 1)(r^o + \delta)$ if $\tau^{ko} = 0$
 - larger markup helps satisfy the condition

Calibration and Simulation

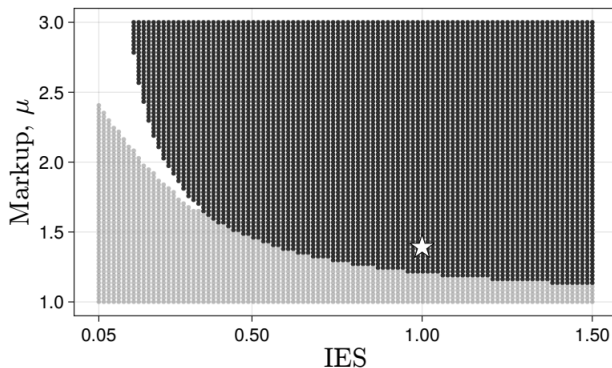


Figure: RPI Feasibility

details

Summary

- introduce rather restrictive but robust welfare metric: RPI
- provide sufficient conditions for RPI to be feasible
- highlight the importance of **aggregate** savings elasticity
- government debt useful in environment with low interest rate or markup

Appendix

- taxes paid by firms $= F(K_t, N^o) - \Pi^o - w^o N^o - (r_t + \delta)K_t$
- initially, tax revenue is purely used to pay interest on debt B^o ($T^o = 0$)
 - $r^o B^o = F(K^o, N^o) - \Pi^o - w^o N^o - (r^o + \delta)K^o$
 - $\Rightarrow -\Pi^o - w^o N^o = r^o B^o - F(K^o, N^o) + (r^o + \delta)K^o$
- combine, and we get
$$TR = F(K_t, N^o) + r^o B^o - F(K^o, N^o) + (r^o + \delta)K^o - (r_t + \delta)K_t$$
- gov BC: $B_{t+1} + TR \geq (1 + r_t)B_t + T_t$
 - $B_{t+1} - (1 + r_t)B_t - T_t \geq -TR = RHS$

back

$$\begin{aligned}C_t &= w^o N^o + \Pi^o + (1 + r_t) \mathcal{A}_t + T_t - \mathcal{A}_{t+1} \\&= \underbrace{F(K^o, N^o) - r^o B^o - (r^o + \delta) K^o}_{w^o N^o + \Pi^o} \\&\quad + (1 + r_t)(K_t + B_t) + T_t - (K_{t+1} + B_{t+1})\end{aligned}$$

- just substitute this into government BC
- we could always (implicitly) set $B_{t+1} = \mathcal{A}_{t+1}(\mathbf{r}, \mathbf{T}) - K_{t+1}$, so long as $\{K_t\}$ exists

Lemma 2

Suppose $K^0 > K^*$ and consider a sequence $\hat{\mathbf{T}} = (T_0, T_1, \dots)$ with $T_t \geq 0 = T^o$ for all $t \geq 0$ with at least one inequality strict. Suppose there exists an interval $(0, \varepsilon)$ and an $M > 0$ for which the following regularity condition holds:

$$\left| C_t(\mathbf{r}^o, \mathbf{T}^o + \hat{\mathbf{T}}_v) - C_t(\mathbf{r}^o, \mathbf{T}^o) \right| \leq Mv, \text{ for all } t \geq 0 \text{ and } v \in (0, \varepsilon)$$

Then there exists an implementable RPI.

back

Appendix

- $\beta = 0.993$
- Epstein-Zin preference: $\gamma = 5.5$
- $r^o = -1.4\%$ (average difference of one-year treasury rate and nominal GDP growth)
- $\tau = 50$ years
- benchmark: markup $\mu = 1.4$, IES $\zeta = 1$

back